

Tailwind CSS

Utility-first CSS framework 

What is Tailwind CSS?

Tailwind CSS is a utility-first CSS framework that helps us build beautiful websites faster by using ready-made classes. 

You write **HTML**
You use **Classes**
You build **Fast!** 

Why Tailwind?

- ☒ No need to write custom CSS
- ☒ Very fast development
- ☒ Fully responsive 
- ☒ Highly customizable  



Tailwind CSS

Rapidly build modern websites without ever leaving your HTML. 

How does it work?

You add utility classes directly in your HTML to style elements. 

Example

```
<h1 class="text-3xl font-bold text-blue-600">  
Hello Tailwind!   
</h1>
```

How to use?

- 1 Add Tailwind CDN in <head>

```
<script src="https://cdn.tailwindcss.com"></script>
```

- 2 Start using classes in your HTML.

```
<button class="bg-blue-500 text-white px-4 py-2  
rounded hover:bg-blue-600">
```

Click Me

```
</button>
```

Note

- No writing long CSS
- Reusable classes
- Clean & consistent design
- Mobile-first approach 

Write less code, do more with Tailwind 



Spacing & Sizing

Tailwind uses a spacing scale based on 0.25rem (4px).

The spacing scale makes it easy to keep consistency in your design!

Spacing Scale

Scale	rem	px (approx.)
0	0	0px
0.5	0.125rem	2px
1	0.25rem	4px
2	0.5rem	8px
3	0.75rem	12px
4	1rem	16px
5	1.25rem	20px
6	1.5rem	24px
8	2rem	32px
10	2.5rem	40px
12	3rem	48px
16	4rem	64px
20	5rem	80px
24	6rem	96px

Margin (m)

Add margin outside an element.

m-0 → margin: 0
 m-1 → margin: 0.25rem (4px)
 m-2 → margin: 0.5rem (8px)
 m-4 → margin: 1rem (16px)
 m-8 → margin: 2rem (32px)
 m-auto → margin: auto

mt-4 → margin-top
 mb-4 → margin-bottom
 ml-4 → margin-left
 mr-4 → margin-right
 mx-4 → margin-left & right
 my-4 → margin-top & bottom

Padding (p)

Add padding inside an element.

p-0 → padding: 0
 p-1 → padding: 0.25rem (4px)
 p-2 → padding: 0.5rem (8px)
 p-4 → padding: 1rem (16px)
 p-8 → padding: 2rem (32px)

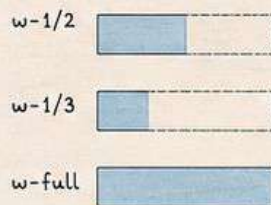
pt-4 → padding-top
 pb-4 → padding-bottom
 pl-4 → padding-left
 pr-4 → padding-right
 px-4 → padding-left & right
 py-4 → padding-top & bottom

Tip: Use m-auto to center block elements horizontally.

Width (w)

w-0 → 0
 w-1/2 → 50%
 w-1/3 → 33.33%
 w-2/3 → 66.66%
 w-1/4 → 25%
 w-3/4 → 75%
 w-full → 100%
 w-screen → 100vw
 w-auto → auto

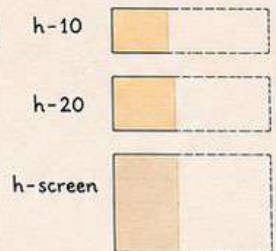
Examples



Height (h)

h-0 → 0
 h-10 → 2.5rem (40px)
 h-20 → 5rem (80px)
 h-screen → 100vh
 h-full → 100%
 h-auto → auto

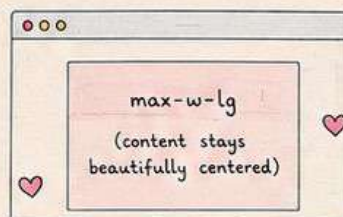
Examples



Max Width (max-w)

max-w-xs → 20rem (320px)
 max-w-sm → 24rem (384px)
 max-w-md → 28rem (448px)
 max-w-lg → 32rem (512px)
 max-w-xl → 36rem (576px)
 max-w-2xl → 42rem (672px)
 max-w-4xl → 56rem (896px)
 max-w-full → 100%

Example



Remember

- Margin is outside
- Padding is inside
- Use spacing scale for consistency
- Combine utilities to build beautiful layouts!

Consistent spacing = Clean & Professional Design

Typography & Colors

Make your text beautiful & your website colorful ♥

Text Size ♥

text-xs	→ extra small
text-sm	→ small
text-base	→ base (default)
text-lg	→ large
text-xl	→ extra large
text-2xl	→ 2x large
text-3xl	→ 3x large
text-4xl	→ 4x large
text-5xl	→ 5x large
text-6xl	→ 6x large
text-7xl	→ 7x large
text-8xl	→ 8x large
text-9xl	→ 9x large

Font Weight ♥

font-thin
font-light
font-normal (default)
font-medium
font-semibold
font-bold
font-extrabold
font-black

Text Alignment ♥

text-left → align left
text-center → center
text-right → align right
text-justify → justify

Font Style

italic → italic text
not-italic → normal text

Example

```
<h1 class="text-3xl font-bold
text-center text-blue-600">
```

Hello Tailwind!

```
</h1>
```

(3xl size, bold, center
& blue text) ♥

Colors in Tailwind

Tailwind has a beautiful default color palette ♥

Text Colors (text-*)

●	text-gray-900
●	text-gray-700
●	text-gray-500
●	text-gray-300
○	text-white
●	text-black
●	text-blue-500
●	text-green-500
●	text-red-500
●	text-yellow-500
●	text-purple-500
●	text-pink-500

Use these
to change
text color
easily!



Background Colors (bg-*)

○	bg-white
●	bg-black
●	bg-gray-100
●	bg-gray-200
●	bg-gray-800
●	bg-blue-500
●	bg-green-500
●	bg-red-500
●	bg-yellow-400
●	bg-purple-500
●	bg-pink-500

Use for
section,
cards,
buttons
& more!



Example

```
<p class="text-gray-700
text-center">
```

This is a paragraph.

```
</p>
```

```
<div class="bg-blue-500
text-white p-4 rounded-lg">
```

Blue Box

```
</div>
```

bg- = background
text = text color
p = padding
rounded = rounded corners

Pro Tip

- Use contrast colors for better readability.

Flexbox, Border, Shadow & Responsive Design

Tailwind makes layouts easy, beautiful and fully responsive!

1 FLEXBOX (LAYOUT)

Use Flexbox utilities to build layouts and align items easily. ♥

flex	→ display: flex
inline-flex	→ display: inline-flex
flex-row	→ row direction (default)
flex-col	→ column direction
items-start	→ align items at start
items-center	→ align items center
items-end	→ align items end
justify-start	→ justify content start
justify-center	→ center content
justify-between	→ space between items
justify-around	→ space around items
gap-2	→ gap: 0.5rem (8px)
gap-4	→ gap: 1rem (16px)
gap-x-4	→ horizontal gap
gap-y-4	→ vertical gap

Examples



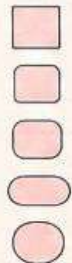
2 BORDERS & RADIUS

Borders

border	→ 1px solid border
border-0	→ no border
border-2	→ 2px solid border
border-t	→ top border
border-b	→ bottom border
border-l	→ left border
border-r	→ right border
border-gray-300	→ gray border
border-dashed	→ dashed border
border-dotted	→ dotted border
border-double	→ double border

Radius (rounded corners)

rounded-none	→ no rounding
rounded-sm	→ small
rounded	→ default
rounded-md	→ medium
rounded-lg	→ large
rounded-xl	→ extra large
rounded-2xl	→ 2x large
rounded-full	→ full (circle)



3 SHADOWS

shadow-sm	→ small shadow
shadow	→ regular shadow
shadow-md	→ medium shadow
shadow-lg	→ large shadow
shadow-xl	→ extra large shadow
shadow-2xl	→ 2x large shadow
shadow-inner	→ inner shadow
shadow-none	→ no shadow

Example



4 HOVER & TRANSITIONS

Hover

hover:bg-blue-500	→ background on hover
hover:text-white	→ text color on hover
hover:scale-105	→ scale on hover
hover:shadow-lg	→ shadow on hover
hover:underline	→ underline on hover
hover:opacity-80	→ opacity on hover

Transitions

transition	→ smooth transition
duration-200	→ 200ms
duration-300	→ 300ms
duration-500	→ 500ms
ease-in-out	→ smooth speed curve

Button Example

```
<button class="bg-blue-500
text-white px-4 py-2 rounded
transition duration-300
hover:bg-blue-600
hover:scale-105">
  Hover Me
</button>
```



5 RESPONSIVE DESIGN (BREAKPOINTS)

Tailwind is mobile-first. Add prefixes to apply styles at different screen sizes.

sm	md	lg	xl	2xl
640px+	768px+	1024px+	1280px+	1536px+

Example

```
<div class="text-center text-sm md:text-base
lg:text-lg xl:text-xl 2xl:text-2xl
p-4 md:p-6 lg:p-8 bg-gray-100 rounded-lg">
  Responsive Text & Padding
</div>
```



Quick Tips

- ✓ Use spacing scale for consistency.
- ✓ Combine utilities to build beautiful UIs.
- ✓ Mobile-first approach.
- ✓ Keep it simple and clean!



Mini Cheat Sheet

Display

block
inline-block
inline
hidden

Flex

flex
inline-flex
flex-col
flex-row

Align

items-center
items-start
items-end
justify-center

Spacing

m-4 (margin)
p-4 (padding)
gap-4 (gap)

Other

rounded-lg
shadow-md
text-gray-700
bg-blue-500

Helpful Stuff

You focus on building, not writing CSS!

1) BACKGROUND

bg-{color}	→ background color
bg-transparent	→ transparent bg
bg-gradient-to-r	→ gradient left to right
bg-cover	→ cover entire area
bg-contain	→ contain without crop
bg-center	→ center the bg
bg-no-repeat	→ don't repeat

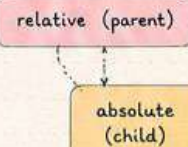
Example

```
<div class="bg-gradient-to-r from-pink-400
to-purple-500 text-white p-6 rounded-lg">
  Gradient Background
</div>
```

2) DISPLAY & POSITION

block	→ display: block
inline	→ display: inline
inline-block	→ inline but accepts width/height
hidden	→ display: none
relative	→ position: relative
absolute	→ position: absolute
fixed	→ position: fixed
sticky	→ position: sticky (on scroll)
z-10 / z-50	→ control stacking order

Example



3) OPACITY & OVERFLOW

opacity-0	→ 0%	overflow-auto	→ scroll if needed
opacity-25	→ 25%	overflow-hidden	→ hide overflow
opacity-50	→ 50%	overflow-x-auto	→ x-axis scroll
opacity-75	→ 75%	overflow-y-auto	→ y-axis scroll
opacity-100	→ 100%		



Example

```
<div class="w-64 h-32 overflow-auto border p-2">
  Long content... Long content... Long content...
</div>
```

4) TRANSFORMS

scale-90	→ scale(0.9)
scale-100	→ scale(1)
scale-110	→ scale(1.1)
rotate-0	→ rotate(0deg)
rotate-12	→ rotate(12deg)
-rotate-12	→ rotate(-12deg)
translate-x-2	→ move right
-translate-x-2	→ move left
translate-y-2	→ move down
-translate-y-2	→ move up

Example (on hover)

```
<button class="bg-blue-500
text-white px-4 py-2
rounded-lg transform
transition duration-300
hover:scale-105
hover:-translate-y-1">
  Hover Me
</button>
```

5) FILTERS

blur-sm	→ small blur
blur-md	→ medium blur
brightness-75	→ darker
brightness-125	→ brighter
contrast-125	→ more contrast
grayscale	→ gray effect
invert	→ invert colors

Normal



Grayscale



6) INTERACTIVITY

hover:	→ on hover
focus:	→ on focus
active:	→ on click
disabled:	→ when disabled
group	→ parent hover
group-hover:	→ child on parent hover
peer	→ sibling state
peer-checked:	→ sibling when checked

Example (hover)

```
<div class="group p-4 border
rounded-lg hover:bg-gray-100">
  <h3 class="group-hover-text-blue-600
font-semibold"> Title </h3>
</div>
```

Hover on box → title color changes!

7) GRID (BASICS)

grid	→ display: grid
grid-cols-1 / 2 / 3 / ...	→ columns
grid-rows-1 / 2 / ...	→ rows
col-span-1 / 2 / ...	→ span columns
row-span-1 / 2 / ...	→ span rows
gap-4	→ gap between items

Example

```
<div class="grid grid-cols-3 gap-4">
  <div class="bg-pink-200 p-4 rounded">1</div>
  <div class="bg-blue-200 p-4 rounded">2</div>
  <div class="bg-green-200 p-4 rounded">3</div>
</div>
```

1

2

3

8) COMMON COLORS (SAMPLE)

gray					
red					
pink					
purple					
blue					
green					
yellow					

Use: bg-color-500
text-color-700 border-color-300

Best Practices

- ♥ Use utilities to build fast.
- ♥ Keep HTML clean & semantic.
- ♥ Combine classes smartly.
- ♥ Use responsive prefixes.
- ♥ Consistent spacing & colors.
- ♥ Test on mobile too!

Advanced Utilities

in Tailwind CSS

Some more powerful classes to make your UI even better! ✨

Remember ✨

- Utility-first
- Reusable
- Customizable
- Fast & Consistent



1 Width / Height (More)

min-w-0	→ min width 0
min-h-0	→ min height 0
max-w-xs	→ max width 20rem (320px)
max-w-sm	→ max width 24rem (384px)
max-w-md	→ max width 28rem (448px)
max-w-lg	→ max width 32rem (512px)
max-w-xl	→ max width 36rem (576px)
max-w-2xl	→ max width 42rem (672px)
max-w-full	→ max width 100%
max-h-screen	→ max height screen
h-screen	→ height: 100vh
w-screen	→ width: 100vw

Example

```
<div class="max-w-lg mx-auto h-screen bg-gray-100"></div>
```

2 Z-Index

z-0	→ z-index: 0
z-10	→ z-index: 10
z-20	→ z-index: 20
z-30	→ z-index: 30
z-40	→ z-index: 40
z-50	→ z-index: 50
z-auto	→ z-index: auto



Example

```
<div class="relative">
  <div class="absolute top-0 left-0 z-10 bg-white p-4">On Top</div>
  <div class="absolute top-2 left-2 z-0 bg-blue-200 p-4">Behind</div>
</div>
```

3 Visibility

visible	→ visibility: visible
invisible	→ visibility: hidden
collapse	→ remove from layout (table elements)

Example

```
<p class="invisible">You can't see me</p>
```

Tip:

Use invisible to hide but keep space, hidden to completely remove! ❤️

4 Object Fit & Position

Object Fit

object-contain	→ contain in box
object-cover	→ cover the box
object-fill	→ fill the box
object-none	→ default
object-scale-down	→ scale down

Object Position

object-top
object-center
object-bottom
object-left
object-right

Example

```

```

5 Cursor & Pointer Events

Cursor

cursor-auto	→ default
cursor-pointer	→ pointer
cursor-not-allowed	→ not allowed
cursor-wait	→ wait
cursor-move	→ move
cursor-text	→ text select
cursor-help	→ help

Pointer Events

pointer-events-auto	→ enabled
pointer-events-none	→ disabled

Example

```
<button class="cursor-pointer hover:opacity-80 pointer-events-auto">Click Me</button>
```

6 Aspect Ratio

aspect-square	→ 1 / 1
aspect-video	→ 16 / 9
aspect-auto	→ auto
aspect-[4/3]	→ custom (4:3)
aspect-[21/9]	→ custom (21:9)



Example

```
<div class="w-full aspect-video bg-gray-200 rounded-xl"></div>
```

7 ! Important & Arbitrary

! Important

Add ! before class to make it important.

Example:

```
<p class="!text-red-500">Important Text</p>
```

Arbitrary Values

Use [] to add custom values.

Examples:

```
<div class="w-[350px] h-[200px] text-[17px] leading-[28px] rounded-[12px] my-[30px] p-[22px] shadow-[0_10px_25px_rgba(0,0,0,0.15)]">Custom Box</div>
```



in Tailwind CSS

perfect.
Build more, learn
more! ✨

1. RESPONSIVE DESIGN

Tailwind is mobile-first. Add breakpoints before a class to apply styles on larger screens.

Breakpoints

sm	≥ 640px
md	≥ 768px
lg	≥ 1024px
xl	≥ 1280px
2xl	≥ 1536px

Example

```
<div class="
  text-sm sm:text-base
  md:text-lg lg:text-xl
  p-2 sm:p-4 md:p-6
  grid grid-cols-1
  md:grid-cols-2 lg:grid-cols-3
">
  ...</div>
```

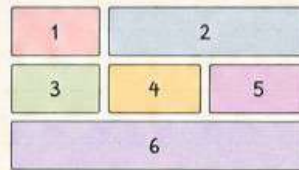
Tip: Write base (mobile) class first, then sm:, md:, lg:...

2. GRID

Use Grid to build powerful 2D layouts.

Basic Classes

grid	→ display: grid
grid-cols-3	→ 3 equal columns
grid-rows-2	→ 2 equal rows
gap-4	→ gap: 1rem (16px)
col-span-2	→ element span 2 cols
row-span-2	→ element span 2 rows
justify-items-center	→ center items
items-center	→ align items center



Example

```
<div class="grid grid-cols-3
  gap-4 p-4">
  <div class="bg-pink-200
    p-4 rounded">1</div>
  <div class="bg-blue-200
    p-4 rounded col-span-2">2</div>
  <div class="bg-green-200
    p-4 rounded">3</div>
  <div class="bg-yellow-200
    p-4 rounded">4</div>
  <div class="bg-purple-200
    p-4 rounded">5</div>
  <div class="bg-indigo-200
    p-4 rounded">6</div>
</div>
```

3. DARK MODE

Add dark: variant to apply styles in dark mode.

How to use?

1. Add 'class' to <html> tag.
2. Add dark: classes.
3. Toggle 'dark' class in JS.

Example

```
<div class="bg-white dark:bg-gray-900
  text-gray-900 dark:text-white
  p-6">
  Dark mode is cool! 🌙
</div>
```

Note: You can also use 'media' strategy in tailwind.config.js for prefers-color-scheme.

4. TRANSITIONS & ANIMATIONS

Transitions

transition	→ all properties
duration-300	→ 300ms
duration-500	→ 500ms
ease-in-out	→ smooth
delay-150	→ 150ms delay

Example

```
<button class="bg-indigo-500 text-white
  px-4 py-2 rounded-lg transition
  duration-300 hover:bg-indigo-600
  ease-in-out">Hover Me</button>
```

Animations (built-in)

animate-spin	→ spin
animate-pulse	→ pulse
animate-bounce	→ bounce
animate-ping	→ ping
animate-fade-in	→ fade in

Example

```
<div class="flex gap-4 items-center">
  <div class="w-8 h-8 bg-pink-500
    rounded-full animate-bounce"></div>
  <div class="w-8 h-8 bg-yellow-500
    rounded-full animate-pulse"></div>
</div>
```

5. FORMS & INPUTS

Useful classes for beautiful & accessible forms.

focus:	→ on focus
placeholder:	→ placeholder text
disabled:	→ disabled state
ring:	→ focus ring
focus:ring-2	→ ring width
focus:ring-blue-500	→ ring color
appearance-none	→ remove default style

Example

```
<input type="text" placeholder="Enter name"
  class="w-full px-4 py-2 border rounded-lg
  focus:outline-none focus:ring-2 focus:ring-blue-500
  placeholder-gray-400 disabled:bg-gray-100" />
```

States

Default

Focus

Disabled

Placeholder

6. PSEUDO-CLASS VARIANTS

Common Variants

hover:	→ on hover
focus:	→ on focus
active:	→ on click
visited:	→ visited links
first:	→ first child
last:	→ last child
nth-odd:	→ odd items
nth-even:	→ even items

Group Hover

Use group to style child on parent hover.

Example

```
<div class="group p-4 bg-gray-100 rounded">
  <h3 class="group-hover:text-blue-600
    transition">Hello</h3>
  <p class="text-gray-600 group-hover:text-gray-900">
    I change on parent hover.</p>
</div>
```

Peer

Use peer to style based on sibling state.

Example

```
<input type="checkbox" id="chk" class="peer hidden" />
<label for="chk" class="peer-checked:bg-green-500
  px-4 py-2 rounded bg-gray-200 cursor-pointer">
  Check Me</label>
```

7. CONTAINER & LAYOUT

container	→ responsive fixed width
mx-auto	→ center horizontally
max-w-screen-lg	→ max width
flex	→ flexible box
justify-center	→ center horizontally
items-center	→ center vertically
min-h-screen	→ full viewport height

Example

```
<div class="min-h-screen
  bg-gray-50 flex
  items-center justify-center">
  <div class="container mx-auto
    max-w-screen-lg p-6">
    Content Here
  </div>
</div>
```

8. CUSTOMIZATION (CONFIG)

Customize Tailwind using tailwind.config.js

What can we customize?

- Colors
- Fonts
- Spacing
- Border Radius
- Screens (Breakpoints)
- Box Shadow

Example (colors)

```
// tailwind.config.js
module.exports = {
  theme: {
    extend: {
      colors: {
        brand: {
          100: '#fce7f3',
          500: '#e91e63',
          700: '#b2182b',
        }
      }
    }
  }
}
```